

Measuring Maritime Disruptions

A route-based index of U.S. import exposure

Summary: We construct a monthly maritime pressure index that measures the exposure of U.S. imports to disruptions at chokepoints such as the Suez Canal and the Strait of Hormuz. The index assigns imports to shipping routes, measures disruptions to vessel traffic at the chokepoints along each route, and weights these disruptions by the value of the imports that use the route. Breaking the index down by product and by U.S. port of entry shows which goods are exposed and where they enter the country. The index peaked in January 2024, about five times its 2022–23 average, when the Red Sea disruption coincided with drought restrictions at the Panama Canal. That episode exposed a broad range of nonenergy goods, which entered mainly through East Coast ports; apparel and footwear contributed more than machinery and electronics. The Hormuz disruption during the 2026 war with Iran was narrower, with pressure falling disproportionately on energy imports and on refining centers along the Gulf and West Coasts.

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1 Introduction

Nearly half of the value of U.S. goods imports, as well as most of their weight, arrives by sea. Much of this trade passes through a handful of narrow waterways, three of which have been disrupted since 2023. Drought forced the Panama Canal to limit transits in 2023. Late that year, attacks on commercial ships in the Red Sea led many carriers to reroute around the Cape of Good Hope, lengthening voyages and raising freight rates. The war with Iran that began at the end of February 2026 then all but halted traffic through the Strait of Hormuz.

Disruptions like these raise the cost of delivering the goods that U.S. firms and households buy from abroad, and they slow deliveries down. A manufacturer can be left waiting for components, or a retailer short of inventory, and the effects can reach firms that never import directly but rely on suppliers who do. How much a given disruption matters depends on which goods and routes it touches and on how much the United States spends on them.

In this article, we construct a monthly maritime pressure index that measures the exposure of U.S. imports to these disruptions since January 2022. We start from Census data on imports by U.S. port, partner country, and product; trace each import flow along its likely sea route to the United States; and use IMF PortWatch data to measure disruptions to vessel traffic at the chokepoints along those routes. The index weights these disruptions by the value of the imports that use each route. Because it is built up from individual flows, the index can be broken down by chokepoint, by product, and by U.S. port of entry.

We measure disruptions by the value of the imports they affect following [Fuchs, Leibovici, and Wong \(2026\)](#), in which we show that, to a first order, the direct cost of a small shipping disruption equals the expenditure-weighted average of the proportional changes in delivery costs it causes. A moderate shortfall on a route that carries a large share of U.S. imports can therefore matter more than a severe shortfall on a route that carries little.

We have three main findings. First, the Red Sea disruption placed more pressure on U.S. import routes than any other episode in our sample, including the Hormuz disruption (Figure 1.1). The index peaked at 38.88 points in January 2024, about five times its 2022–23 average, when the start of the Red Sea disruption coincided with drought restrictions at the Panama Canal, and it remained elevated through late 2025.



Figure 1.1. The national maritime pressure index

January 2022–August 2026. Index points are 100 times the vessel-import-weighted average route log shortfall, using the preceding three calendar years’ trade values. Dashed lines mark January 2024 and March 2026. Sources: U.S. Census Bureau, SeaRoute, IMF PortWatch; authors’ calculations.

Second, the Red Sea disruption reached a broad range of nonenergy goods, whereas the Hormuz disruption was concentrated in energy imports. In January 2024, apparel and footwear contributed more to the index than machinery and electronics, even though they account for only about half as much maritime trade.

Third, the goods exposed to the Red Sea disruption entered the country mainly through East Coast ports, led by New York/New Jersey, Savannah, and the Port of Virginia. Los Angeles/Long Beach, which handles more maritime import value than any other U.S. port, contributed little, because its

imports cross the Pacific on routes that avoid the Red Sea. During the Hormuz disruption, pressure fell disproportionately on refining centers along the Gulf and West Coasts.

Because it relies on public data that are updated continually, the index can serve as a monitoring tool. It can be recomputed as events unfold, and its breakdowns identify which goods and ports are exposed while a disruption is under way, before the affected goods reach U.S. ports and well before the disruption appears in official trade statistics. The same exposure measures can also be used to assess a hypothetical disruption at a chokepoint that has not yet been affected. The recent episodes also show that a disruption can move the national index only modestly, as the Hormuz disruption did, and still put severe pressure on particular goods and on the ports that receive them.

Our index complements existing measures of supply-chain stress, such as the Global Supply Chain Pressure Index of [Benigno et al. \(2022\)](#), which combines transportation costs with manufacturing survey data, and IMF PortWatch, which reports vessel activity at ports and chokepoints around the world ([International Monetary Fund and University of Oxford 2023](#)). We add a link between specific chokepoints and U.S. imports, by product and port of entry, grounded in economic theory. Recent models of the global shipping market study how disruptions such as the Red Sea attacks propagate to shipping costs, trade, and aggregate activity ([Dunn and Leibovici 2024](#); [Brancaccio et al. 2026](#)).

Closely related work by [Bai et al. \(2026\)](#) combines detailed vessel-tracking data with a model of U.S.-bound shipping to explain how disruptions spread through the economy. Their analysis shows how shipping-capacity constraints and congestion, together with retailers' sourcing, sales, and entry decisions, can amplify the effects of a disruption. We complement their analysis by using publicly available data to construct an index of exposure to maritime chokepoints across U.S. ports and product categories.

An empirical literature documents why this exposure is costly, estimating that each additional day in transit is equivalent to an ad valorem tariff of 0.6 to 2.1 percent ([Hummels and Schaur 2013](#)), that higher shipping costs pass through to consumer prices ([Carrière-Swallow et al. 2023](#)), and that shortages and delivery delays are associated with higher retail prices ([Cavallo and Kryvtsov 2023](#); [Baslandze and Fuchs 2026](#)).

Section 2 presents the economic framework behind the index, Section 3 describes the data and how we build it, Sections 4 and 5 examine differences across goods and across ports, and Section 6 concludes. Appendix A provides the construction details.

2 Analytical Framework

One important way shipping disruptions affect the United States is by raising the cost of delivering the goods it imports. Measuring U.S. exposure to these disruptions therefore requires a way to add up disruptions across many goods, trading partners, and routes. This section lays out a simple framework for doing so, which guides how we build the index in Section 3.

The answer is familiar from price indexes. A 10 percent increase in the price of a good matters more to a household the larger the share of its budget spent on that good, and delivery costs work

the same way. A disruption that raises the delivered price of an import flow by 1 percent matters more, the more is spent on that flow, so changes in delivery costs should be added up in proportion to spending on the affected imports.

In [Fuchs, Leibovici, and Wong \(2026\)](#), we formalize this idea in a model in which shipping services link producers to buyers. Write the delivered price of import flow i as the producer's price times a delivery-cost factor τ_i , and let s_i be the flow's share of baseline import spending. For a small disruption that changes delivery costs by $d \ln \tau_i$, the direct delivery-cost effect on imported purchases, measured relative to import spending, is

$$\sum_i s_i d \ln \tau_i. \quad (2.1)$$

Equation (2.1) makes this precise. Adjustments such as rerouting or switching suppliers do not need to be counted separately. They shape how much delivery costs change, but once those changes are known, baseline spending is enough to value them, to a first order.¹

Our index applies this logic to what we can observe, using disruptions to vessel traffic along each import's route in place of changes in delivery costs and Census import values in place of spending. If a given disruption to traffic raises delivery costs by the same proportion on every route, the index moves roughly in proportion to the direct cost in equation (2.1). The index captures only this delivery-cost channel, however. A disruption can also raise the prices that foreign producers charge, even for goods that never pass through the affected chokepoint. Oil is the leading example, since a disruption at Hormuz can raise U.S. fuel prices through the world oil price even if the United States imports little oil through the strait.

3 Data and Index Construction

Building the index requires three pieces of data, drawn from the sources listed in Table 3.1: the value of each import flow, the route it travels, and conditions along that route. To make the steps concrete, we follow a single flow through the construction of the index, cotton shirts from India that enter the United States through the Port of Savannah.

Table 3.1. Data used to construct the index

Source	Information	Role
U.S. Census	Imports by U.S. port, partner, product, and mode	Value of each import flow
SeaRoute	Sea routes between foreign and U.S. ports	Chokepoints along each route
IMF PortWatch	Daily vessel traffic at ports and chokepoints	Loading ports and traffic disruptions

¹ This is the logic behind [Hulten's \(1978\)](#) theorem. Like Hulten's result, it assumes an efficient baseline and holds for small disruptions. For large ones, such as the prolonged closure of a major route, higher-order effects can matter ([Baqae and Farhi 2019](#)).

Import Exposure and Routes

Census import records report, for each U.S. port, partner country, and six-digit Harmonized System (HS6) product, the customs value of imports and how they were shipped ([U.S. Census Bureau n.d.](#)). We keep goods that arrive by vessel, which accounted for 44.8 percent of the value of U.S. imports in 2019. An observation is a trade flow that may bundle many shipments; our shirts are one such flow. Census also reports how much of each flow arrived in containers, as apparel typically does, and we treat the containerized part and the rest as separate flows.

Each flow counts in proportion to its average import value over the preceding three calendar years, which we update once a year. The index for 2024 uses average values for 2021–23, for example, and the index for 2026 uses 2023–25. Updating keeps the index in step with U.S. sourcing, which has shifted considerably since 2019, and averaging over three years smooths out year-to-year noise. Because these values come from past years and are fixed within the year, a disruption cannot immediately shrink its own importance in the index by cutting the flow of goods.

We hold the 2019 route mix fixed within each U.S.-port-partner-vessel-class group and apply it to all products in that group, including newer products. This lets the trade basket change without using post-disruption voyages to redraw exposure. Imports in relationships with no baseline route allocation remain in the denominator but contribute no measured pressure.

Census does not record the foreign port where goods are loaded. We therefore allocate each partner’s shipments to the United States across its ports in proportion to their 2019 export activity for the relevant vessel class, as measured by PortWatch. India’s container trade, for instance, is spread across several Indian ports, including Nhava Sheva, near Mumbai. We then trace the route from each foreign port to the U.S. port of entry using SeaRoute, an open-source tool from Eurostat that computes the shortest path by sea between two ports along a global network of shipping lanes ([Eurostat 2026](#)). The route tells us which chokepoints a shipment would cross on its way to the United States.² The path from Nhava Sheva to Savannah crosses three of the chokepoints we monitor, which we also refer to as gates: Bab el-Mandeb, Suez, and Gibraltar.

Figure 3.1 maps the resulting network of likely routes, which actual voyages need not follow exactly. Line widths increase with the value of U.S. imports carried on each route in 2019, and circles show the value of U.S. imports crossing each chokepoint. U.S.-bound trade is spread very unevenly across chokepoints: the Panama Canal carries the most, followed by Gibraltar, and Hormuz the least. A major corridor runs from Asia and the Middle East to the East Coast through Bab el-Mandeb, Suez, and Gibraltar. Much of the trade that crosses the Pacific to the West Coast passes no chokepoint at all, and in the 2019 baseline only about half of maritime import value was assigned to routes crossing at least one.

Unlike the import values, which we update every year, the routes do not change over time. They are the shortest sea paths between ports, with foreign ports assigned using 2019 data. If routes instead followed where ships actually sail, trade that avoided the Red Sea by going around the Cape of Good Hope would have been reassigned to a route that crosses no disrupted chokepoint,

² We use SeaRoute’s 5-kilometer network and exclude Arctic passages.

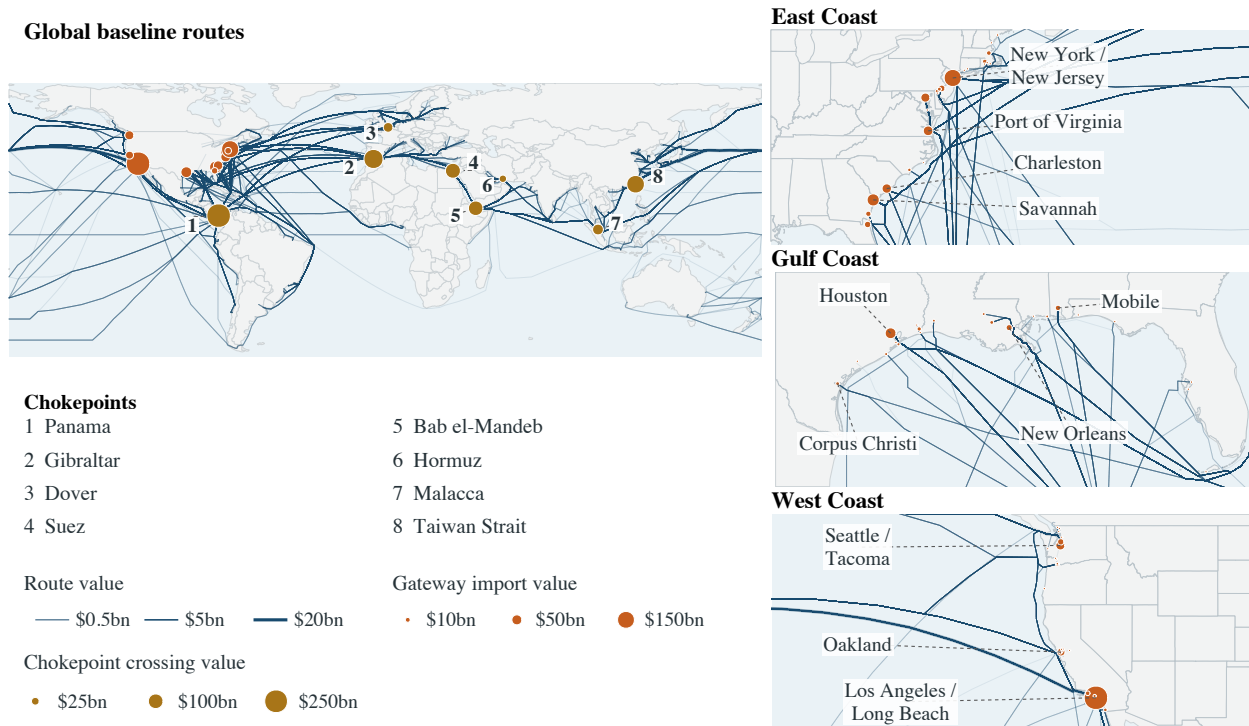


Figure 3.1. Baseline routes locate U.S. import exposure

2019 baseline. Routes shown cover at least 90 percent of assigned vessel-import value. The global map marks the 12 largest U.S. gateways; coastal panels show routes ending on each coast and selected port labels. Line widths share a route-value scale. Orange circle areas show assigned gateway import value on a common scale across panels; gold circle areas show chokepoint-crossing value on a separate scale. Gate values overlap when a route crosses several chokepoints. Sources: U.S. Census Bureau, SeaRoute, IMF PortWatch; authors' calculations.

and the index would have registered little pressure just when the disruption was most severe. Holding routes fixed keeps that trade counted as exposed, which we think appropriate because rerouting is itself a costly response to the disruption.

Measuring Disruptions

We monitor eight chokepoints: Bab el-Mandeb, Dover, Gibraltar, Hormuz, Malacca, Panama, Suez, and the Taiwan Strait. We measure traffic separately for container ships and other vessels and match each flow to its Census shipping class, so that a drop in noncontainer traffic does not register as pressure on containerized goods. The other-vessel class includes tankers but is not a tanker-only measure. At each chokepoint, we compare daily traffic with its normal level for that time of year, defined as the typical level in the same calendar month over the preceding three years. The daily shortfall is the log of normal traffic relative to actual traffic when traffic is below normal, and zero otherwise. The monthly shortfall is the average of these daily values. Appendix A explains the capacity measures and the cap used when traffic nearly stops.

A route's pressure is the largest monthly shortfall among the chokepoints it crosses. We use the maximum rather than the sum because a single interruption often shows up at several chokepoints on the same route. When ships avoid the Red Sea, for example, traffic falls at both Bab el-Mandeb and Suez, and adding the two would count the same interruption twice. For our

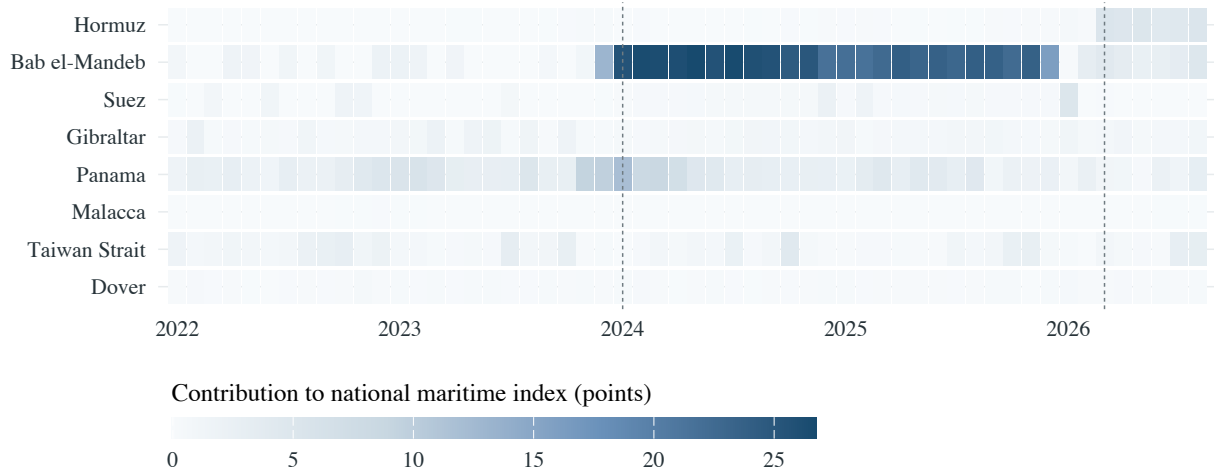


Figure 3.2. Chokepoint contributions to maritime pressure

January 2022–August 2026. Shading shows each chokepoint’s monthly contribution to the national index, in points, using the same annual trade basket as Figure 1.1. A route’s pressure is attributed to its most disrupted gate; the eight contributions sum to the national reading. Sources: U.S. Census Bureau, SeaRoute, IMF PortWatch; authors’ calculations.

shirts, which travel in containers, route pressure in a given month is the largest of the shortfalls in container traffic at Bab el-Mandeb, Suez, and Gibraltar.

Figure 3.2 breaks the index down by chokepoint, with chokepoints in rows and months in columns. The shading shows each chokepoint’s contribution to the national index; each route is assigned to the chokepoint where its shortfall was largest that month. Bab el-Mandeb accounts for most of the pressure from January 2024 through late 2025. Panama, where drought restrictions limited transits at the time, accounts for most of the November 2023 reading and 30.8 percent of the January 2024 reading. The January peak thus reflects two disruptions at once. Hormuz accounts for 40.9 percent of the March 2026 reading. Because each route is assigned to a single chokepoint, the Suez row stays pale during the Red Sea episode even though traffic through Suez fell: on routes that cross both chokepoints, the shortfall at Bab el-Mandeb was larger.

From Route Pressure to the Index

The last step aggregates over import flows, first averaging pressure across each flow’s routes using its route shares and then averaging across flows using their shares of maritime import value. If P_{it} is the pressure assigned to flow i in month t and $w_{i,y(t)}$ is its share of the preceding-three-year vessel-import basket for reporting year $y(t)$, the index is

$$I_t = 100 \sum_i w_{i,y(t)} P_{it}. \quad (3.1)$$

The factor of 100 expresses the index in points, and equation (3.1) is the empirical counterpart of equation (2.1) in Section 2.

To see the scale, suppose routes like the one carrying our shirts account for 10 percent of maritime import value, and traffic at their most disrupted chokepoint runs at 90.5 percent of

normal throughout the month. The log shortfall is then about 0.1, and these routes contribute $100 \times 0.10 \times 0.1 = 1$ point to the index. If twice as much trade used those routes, their contribution would double even though conditions along them were the same.

The denominator includes all vessel imports. The unassigned share is 5.1 percent of the 2024 basket and 5.6 percent of the 2026 basket; it contributes zero, as do routes that cross no monitored chokepoint. Our sample ends in August 2026, the last complete month of PortWatch data available on October 1. The index records conditions at chokepoints in the month they occur, before the affected goods complete their remaining voyages to the United States.

Because the normal level of traffic is based on the preceding three years, a long disruption eventually becomes part of the benchmark. The normal is a median over those three years, so it falls once two of them are disrupted, which for the Red Sea episode happened around the turn of 2026. The sharp drop in the index between November 2025 and January 2026 coincides with this change and, on its own, is not evidence that traffic returned to its pre-2024 level.

4 Differences across Goods

The location of a disruption determines which goods it affects, because different goods travel on different parts of the shipping network. Energy is the clearest case, since crude oil and refined products move on tankers, often along routes that container ships do not use.

Figure 4.1 compares the index for energy imports (HS chapter 27) with the index for all other imports. Each series is computed within the group's own maritime imports and thus measures how severe conditions were for that group, and both panels use the same scale.

Maritime pressure on nonenergy imports rose more than fivefold between October 2023 and January 2024 and remained elevated through late 2025. Maritime pressure on energy imports also rose during the Red Sea episode, but its largest increase came in March 2026, when it rose about ninefold in a single month, and it has remained near that level since. By then, pressure on nonenergy imports had fallen to about a quarter of its January 2024 level.

Figure 4.1 shows how severe conditions were for energy and nonenergy imports, but how much a group of goods matters for the national index also depends on its share of maritime trade.

Figure 4.2 combines the two by breaking the national index into the contributions of eight product groups in January 2024 and March 2026, with each bar measured in points of the national index.

In January 2024, contributions were spread across many groups. Apparel and footwear contributed the most, 7.73 points, closely followed by machinery and electronics at 7.19 points. Machinery and electronics account for more than twice as much maritime trade in that year's basket, but route pressure on apparel and footwear was more than twice as severe. Chemicals, food, transport equipment, and materials also made substantial contributions.

In March 2026, energy was by far the largest contributor, at 4.19 points, while no other group contributed more than 2 points. Energy accounted for 32.8 percent of the March reading, compared with 5.8 percent in January 2024, even though it makes up only 9.4 percent of the 2026 maritime import basket. The national reading was nonetheless much lower than in January 2024.

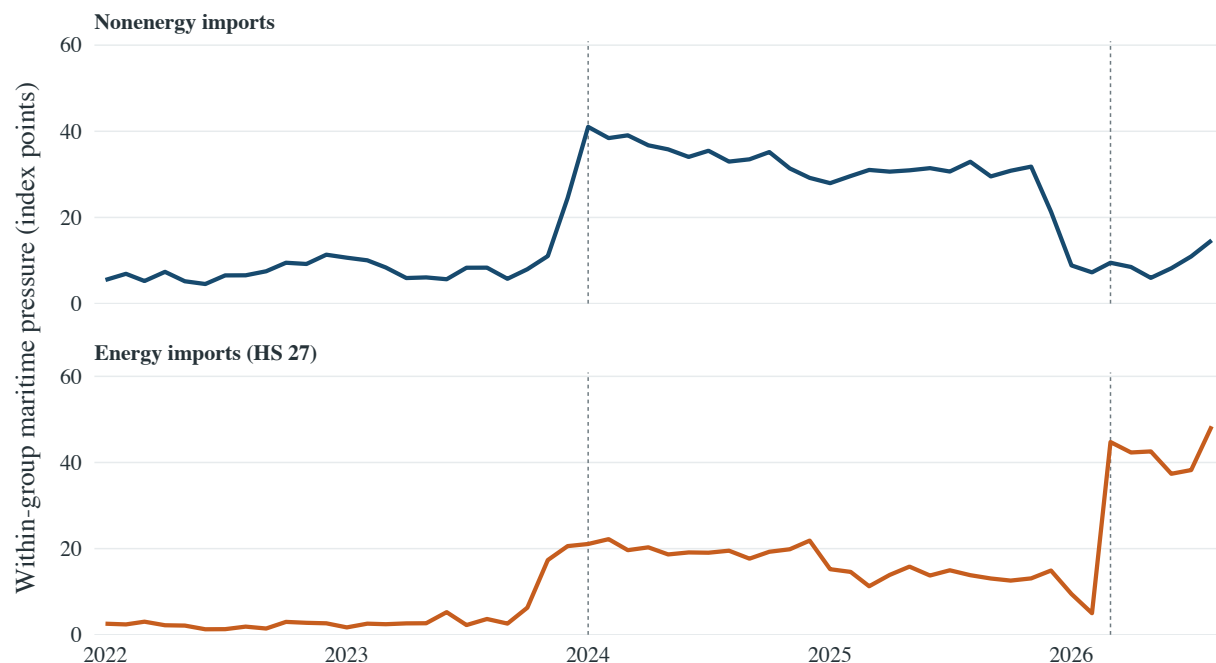


Figure 4.1. Maritime pressure on energy and nonenergy imports

January 2022–August 2026. Each series divides its pressure numerator by the group’s full three-year vessel-import basket, including unassigned imports. The panels share a scale. Dashed lines mark January 2024 and March 2026. Sources: U.S. Census Bureau, SeaRoute, IMF PortWatch; authors’ calculations.

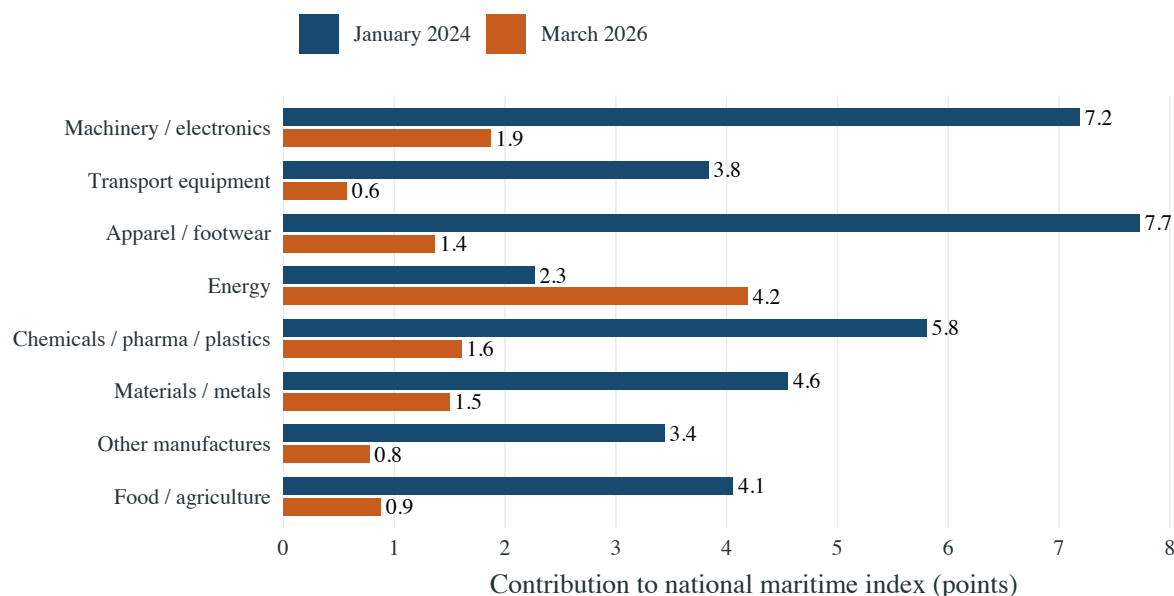


Figure 4.2. Product contributions to national maritime pressure, January 2024 and March 2026

Each bar shows a product group’s contribution in national index points, using the annual basket for that month. The eight unrounded bars sum to 38.88 points in January 2024 and 12.78 in March 2026. Sources: U.S. Census Bureau, SeaRoute, IMF PortWatch; authors’ calculations.

Hormuz carries relatively little U.S.-bound trade in the baseline network (Figure 3.1), so even severe pressure there adds relatively few points, and by March 2026 the contributions of Bab el-Mandeb and Panama had fallen well below their January 2024 levels (Figure 3.2).

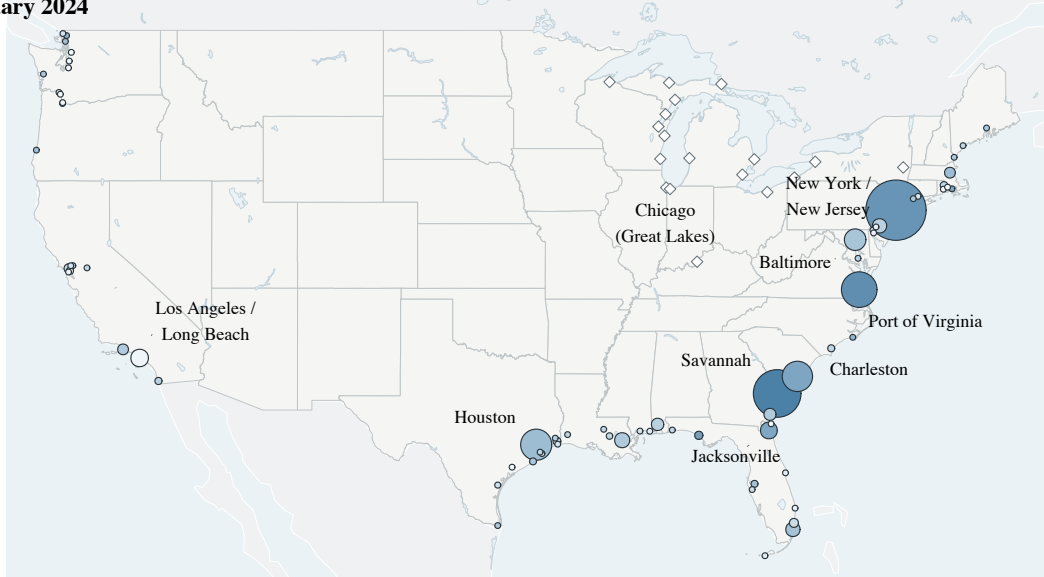
5 Differences across U.S. Ports

Exposure also differs across the U.S. ports through which imports enter the country. Figure 5.1 groups maritime pressure by port of entry, or gateway. Circle area represents the port's contribution to the national index, with a minimum size for visibility; color represents average pressure on its assigned routes. Both panels use the same scales. A small port can face severe pressure and still contribute little, because few imports are exposed. These measures describe pressure on the routes into each port, not congestion at the port itself, and they do not show where the goods are ultimately used, since goods entering Savannah, for example, serve buyers well beyond Georgia.

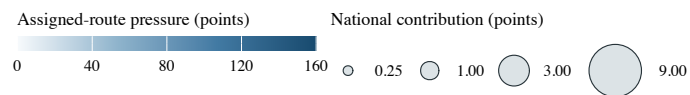
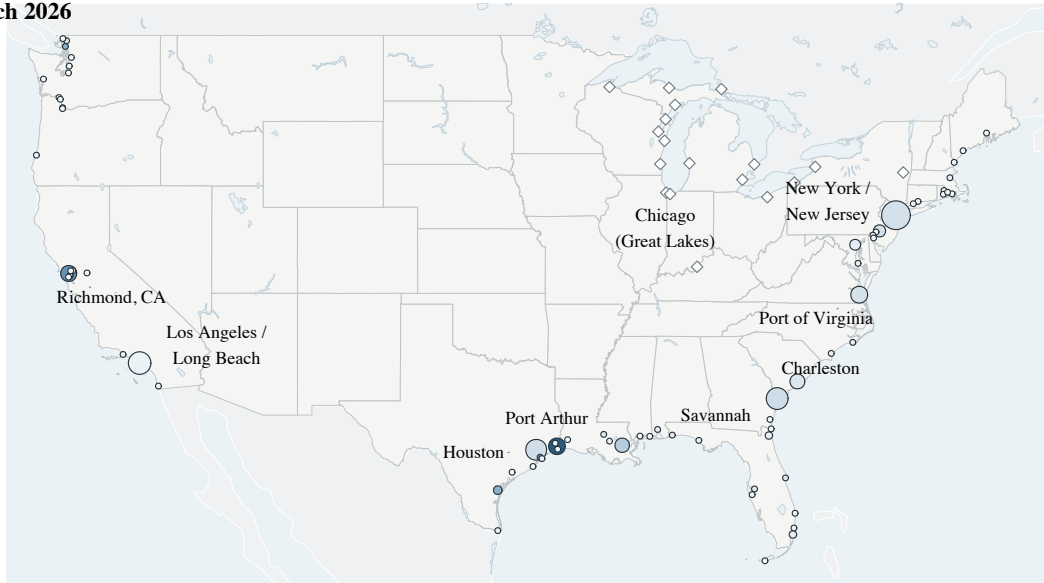
In January 2024, the largest contributions came from the East Coast. New York/New Jersey contributed the most, followed by Savannah and the Port of Virginia, each combining large import values with substantial pressure on routes through the Red Sea and the Panama Canal. Los Angeles/Long Beach, the largest U.S. port of entry for maritime imports, contributed much less, because its imports arrive mostly across the Pacific on routes that avoid the chokepoints disrupted that month.

By March 2026, the East Coast circles had shrunk considerably. Maritime pressure instead fell disproportionately on refining centers along the Gulf and West Coasts, such as Port Arthur, Texas, and Richmond, California, consistent with the concentration of the Hormuz disruption in energy imports. These ports had some of the darkest markers, and their contributions were large relative to the imports they receive, although still smaller than those of New York/New Jersey and Los Angeles/Long Beach, which faced much less pressure but receive far more imports.

A. January 2024



B. March 2026



Circles below 0.074 points share a minimum size; gray crosses denote zero contribution.

Figure 5.1. Maritime pressure by U.S. port of entry, January 2024 and March 2026

Circle area shows contribution to the national index using each month's annual trade basket. Positive contributions below 0.074 points share a minimum size; gray crosses mark zero contributions. Color shows 100 times the value-weighted log shortfall among assigned routes into the gateway. Both months share area and color scales. Gray diamonds mark 18 Great Lakes and inland gateways; eight noncontiguous gateways lie outside the map. Sources: U.S. Census Bureau, SeaRoute, IMF PortWatch; geography: Census and Natural Earth.

6 Concluding Remarks

We have constructed a monthly index of the exposure of U.S. imports to maritime disruptions. Built from public data, the index measures each disruption by the value of the imports it affects, and it can be broken down by chokepoint, product, and port of entry. These breakdowns show how different the two recent episodes were.

The Red Sea disruption, compounded in late 2023 and early 2024 by drought restrictions at the Panama Canal, produced the highest readings in our sample and affected many kinds of nonenergy goods, which entered mainly through East Coast ports. The Hormuz disruption of 2026 was narrower, falling disproportionately on energy imports and on the refining centers that receive them.

Because the index relies only on public data, it can be updated every month to show, while a disruption is still unfolding, which U.S. imports are exposed and where they enter the country. More broadly, the same approach can be applied to any country with similarly detailed import data, since the traffic data and shipping routes it relies on are global.

A Construction Details

Data. The index covers January 2022–August 2026. Import weights come from monthly Census files of imports by U.S. port, partner country, and HS6 product for 2019–25 ([U.S. Census Bureau n.d.](#)). Foreign loading ports are assigned using 2019 PortWatch export activity ([International Monetary Fund and University of Oxford 2023](#)), and routes are computed on SeaRoute’s 5-kilometer network, excluding Arctic passages ([Eurostat 2026](#)).

Weights and routes. A flow i is a U.S. port, partner country, HS6 product, and vessel class (containerized or other). Its weight $\bar{V}_{iy}$ is its average annual customs value over the three calendar years before reporting year y , counting years with no imports as zero. Route shares are fixed at their 2019 values and pooled across products within each U.S.-port-partner-vessel-class group g , so products first imported after 2019 inherit their group’s route mix. We write π_{gr}^{2019} for the share of group g ’s total 2019 vessel-import value assigned to routes crossing the set of chokepoints r . Incomplete or unavailable 2019 route assignments leave 4.5 to 5.6 percent of the basket unassigned, depending on the year. This trade stays in the denominator and contributes zero pressure.

Index. Let h_{bvt} be the monthly shortfall at chokepoint b for vessel class v , and let q_{rvt} be the largest h_{bvt} among the chokepoints in r , with $q_{rvt} = 0$ for routes that cross no monitored chokepoint. With $g(i)$ and $v(i)$ denoting flow i ’s routing group and vessel class, and $y(t)$ the reporting year of month t , the index is

$$I_t = 100 \frac{\sum_i \bar{V}_{i,y(t)} \sum_r \pi_{g(i)r}^{2019} q_{r,v(i),t}}{\sum_i \bar{V}_{i,y(t)}}. \quad (\text{A.1})$$

Chokepoint contributions assign each route’s pressure to the chokepoint with the largest shortfall; product and port contributions restrict the numerator to the relevant flows. Each decomposition keeps the national denominator, so its contributions sum to I_t . The energy and nonenergy series in Figure 4.1 instead divide by each group’s own vessel-import basket, and the colors in Figure 5.1 divide by each port’s routed imports.

Shortfalls. Traffic is PortWatch transit capacity for container ships and other vessels, with vessel counts used where capacity is missing. Normal traffic is the median of the preceding three years’ median daily traffic in the same calendar month. Below normal, the daily shortfall is the log of normal divided by actual traffic; otherwise it is zero. Traffic at or below 4 percent of normal receives the maximum shortfall, $\ln 25 \approx 3.2$. Monthly shortfalls average the daily values.

Interpretation. The weights in equation (2.1) are shares of spending at delivered prices. Census customs values exclude international freight and insurance, so they approximate these shares when freight and insurance are a similar fraction of value across flows. Relating the index to this delivery-cost effect requires a common conversion from traffic shortfalls to cost changes. We do not estimate that conversion, so the index is in traffic-shortfall units.

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